

Stanford CS 221 Sentiment Analysis (SP 2024).

Problem 1:

a: $w \leftarrow w - \eta \nabla_w \text{Loss}_{\text{hinge}}(x, y, w)$

Review one: $\phi(x) = [0, 1, 0, 0, 1, 0]$ Loss = 1

$$\nabla \text{Loss}(x, y, w) = [0, 1, 0, 0, 1, 0]$$

$$w \rightarrow [0, -0.1, 0, 0, -0.1, 0]$$

Two: $\phi(x) = [1, 0, 1, 0, 0, 0]$ Loss = 1

$$\nabla \text{Loss}(x, y, w) = [1, 0, 1, 0, 0, 0]$$

$$w \rightarrow [-0.1, -0.1, -0.1, 0, -0.1, 0]$$

Three: $\phi(x) = [0, 1, 0, 1, 0, 0]$ Loss = 0.9

$$\nabla \text{Loss}(x, y, w) = [0, -1, 0, -1, 0, 0]$$

$$w \rightarrow [-0.1, 0, -0.1, 0.1, -0.1, 0]$$

Four: $\phi(x) = [1, 0, 0, 0, 0, 1]$ $\nabla \text{Loss}(x, y, w) = [-1, 0, 0, 0, 0, 0]$

$$\Rightarrow w \rightarrow [0, 0, -0.1, 0.1, -0.1, 0.1]$$

b : (1)

If we denote the feature vector $\phi(x)$ as counting the occurrences of the words "bad," "good," and "not," we can represent them as follows:

- "bad": $[1, 0, 0] \rightarrow (-1)$
- "good": $[0, 1, 0] \rightarrow (+1)$
- "not bad": $[1, 0, 1] \rightarrow (+1)$
- "not good": $[0, 1, 1] \rightarrow (-1)$

A linear classifier $f_w(x) = \text{sign}(w \cdot \phi(x))$ requires that positive examples yield a positive value and negative examples yield a negative value. The crucial conflict arises between "not bad" (expected to be +1) and "not good" (expected to be -1).

For any linear decision boundary, the vector for "not bad" $[1, 0, 1]$ and "not good" $[0, 1, 1]$ cannot be separated from each other with respect to their assigned labels due to the influence of the negation. Thus, a linear classifier cannot correctly classify all reviews, proving that no linear classifier can achieve zero error on this dataset.

(2):

Proposed Feature

To resolve this issue, we can introduce an additional feature: "**negation indicator**." This feature would represent whether the review contains the word "not" as a separate binary feature (1 for presence, 0 for absence). By including this feature, the classifier can better distinguish between the sentiment of phrases like "not bad" and "not good," allowing it to correctly classify all reviews in the dataset.

Problem 2:

a:
$$\text{Loss}(x, y, w) = [\sigma(w \cdot \phi(x)) - y]^2$$

b:
$$\begin{aligned} p &= \sigma(w \cdot \phi(x)) \Rightarrow \nabla_w \text{Loss}(x, y, w) \quad (p = w \cdot \phi(x)) \\ &= 2(p - y) \frac{\partial \sigma(p)}{\partial p} \frac{\partial p}{\partial w} \\ &= 2(p - y) \phi(x) \cdot (-1) \frac{e^{-p}}{(1 + e^{-p})^2} \sim p(1 - p) \\ &= 2(p - y) p(1 - p) \phi(x). \end{aligned}$$

C :

To minimize the magnitude of the gradient of the loss with respect to w for a single data point (x, y) with $y = 1$, the condition is that the score $w \cdot \phi(x)$ needs to be sufficiently large and positive. Specifically, if w is made arbitrarily large while keeping the direction aligned with $\phi(x)$, the gradient will approach zero.

However, the magnitude of the gradient can never be exactly zero unless $w \cdot \phi(x)$ is exactly equal to one (the threshold for the hinge loss), which typically requires specific values for both w and $\phi(x)$. If w is sufficiently large, the gradient can be made arbitrarily small, but not precisely zero, unless in a specific case.

P3 : 略 .

P4 :
a: $w = [-0.1, 1, 0]$
 $\phi(x) = [1, d, t]$

If-and-Only-If Statements

1. Classifier D:

- $f_D(x) = 1$ if and only if $d = 1$ (demographic identity is present) and t can be either 0 or 1. Gives -1 for all other.

2. Classifier T:

- $f_T(x) = 1$ if and only if $t = 1$ (toxic words are present). Gives -1 for all other.

(1)
b: Avg Loss $f_D(x) = \frac{1}{200} [7+3+3+7] = \frac{1}{10}$.

(2)

	$y = 1$	$y = -1$
$d = 1$	0	1
$d = 0$	1	0

(3): 1 (?)

c: (1): Avg loss $f_T(x) = \frac{1}{200} [27+3+3+27] = 0.3$

	$y = 1$	$y = -1$	
$d = 1$	0.3	0.3	$\Rightarrow 0.3$
$d = 0$	0.3	0.3	max group loss

d:
$$\text{Loss} = \lambda \cdot \text{Loss}_{\text{arg}}(w) + (1-\lambda) \cdot \text{Loss}_{\text{max}}(w), \lambda \in [0,1].$$

Prob 5: $\phi(x_1) = [0, 0]$; $\phi(x_2) = [4, 0]$; $\phi(x_3) = [6, 0]$
 $\phi(x_4) = [11, 0]$

a: (1) $\mu_1 = [0, 0]$ $\mu_2 = [11, 0]$

$$z(x_1) = 1 \quad z(x_2) = 1 \quad z(x_3) = 2 \quad z(x_4) = 2$$

$$\Rightarrow \mu_1 = [2, 0] \quad \mu_2 = \left[\frac{17}{2}, 0\right]$$

loss:

$$L = 2^2 + 2^2 + \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2 = \frac{25}{2} + 8 = \frac{41}{2}$$

(2) $\mu_1 = [0, 0]$ $\mu_2 = [4, 0]$

$$z(x_1) = 1 \quad z(x_2) = 2 \quad z(x_3) = 2 \quad z(x_4) = 2$$

$$\mu_1 = [0, 0] \quad \mu_2 = [7, 0]$$

$$L = 3^2 + 1^2 + 4^2 = 26.$$

C:

1. Scaling All Dimensions:

Response: Yes, scaling all dimensions by the same factor will not change the clustering results in k-means.

Explanation: K-means clustering is based on the distances between data points and the centroids of clusters. When all dimensions are scaled by the same factor, the relative distances between the data points remain unchanged. Since k-means relies on the distances in the same feature space, the assignment of points to clusters will be the same before and after scaling. Therefore, the same data points will belong to the same clusters.

Counter Ex: $\begin{array}{cc} \mu_1 & \mu_2 \\ (1, 1) & (2, 5) \end{array} \xrightarrow{y} \begin{array}{cc} \mu_1' & \mu_2' \\ (11, 0.1) & (12, 0.5) \end{array}$

A (1, 1) B (1, 5) $\times 0.1$ (1, 0.1) (1, 0.4)

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

μ_1 μ_2 μ_1' μ_2'